

CS1231S Discrete Structures

Midterm Test — Answer Sheets

AY2025/26 Semester 1

Time allowed: 1 hour 30 minutes

INSTRUCTIONS

1. Write your **Student Number** on the right AND, using a **pencil** (2B or above), shade the corresponding circle **completely** in the grid for each digit or letter. **DO NOT WRITE YOUR NAME!**
2. Zero mark will be given if you write/shade your Student Number incompletely or incorrectly.
3. Write your Student Number at the top of page 3.
4. There are **FOUR (4) pages** in the Answer Sheets.
5. All questions must be answered in the space provided; no extra sheets will be accepted as answers.
6. You must submit only these **ANSWER SHEETS** and no other documents.
7. An excerpt of the question may be provided to aid you in answering in the correct box. It is not the exact question. You should still refer to the original question in the question paper.
8. You may write your answers in pencil (2B or above) or pen. Pencil is preferred in case you need to erase and rewrite your answers.
9. The maximum mark for this paper is 50.
10. **Marks may be deducted** for (i) illegible handwriting, and/or (ii) excessively long answer.
11. Each multiple choice question is intended to have only one answer. Please shade the appropriate bubble in **pencil**.

STUDENT NUMBER														
A														
U	☐	0	0	0	0	0	0	0	0	A	☐	N	☐	
A	☒	1	1	1	1	1	1	1	1	B	☐	R	☐	
HT	☐	2	2	2	2	2	2	2	2	E	☐	U	☐	
NT	☐	3	3	3	3	3	3	3	3	H	☐	W	☐	
		4	4	4	4	4	4	4	4	J	☐	X	☐	
		5	5	5	5	5	5	5	5	L	☐	Y	☐	
		6	6	6	6	6	6	6	6	M	☐			
		7	7	7	7	7	7	7	7					
		8	8	8	8	8	8	8	8					
		9	9	9	9	9	9	9	9					

For Examiner's Use Only		
Question	Marks	Remarks
Q1-15	/ 30	
Q16	/ 6	
Q17	/ 14	
Total	/ 50	



Any feeling about this midterm test?

This will not be graded! 😊

This page is intentionally left blank.

This page is for your rough work. Whatever is written on this page will NOT be graded.

Explanation for MCQs

Q4: (C) Equals to the number of partitions:

$\{\{1\}, \{2\}, \{3\}\}, \{\{1\}, \{2,3\}\}, \{\{2\}, \{1,3\}\}, \{\{3\}, \{1,2\}\}$ and $\{\{1,2,3\}\}$.

Q5. (C).

(i) every female reader reads a Mystery book and (iii) Mrs Duett reads all Mystery books while at the same time all male readers read at least one Mystery book. Furthermore, no male reader reads all books of a genre and thus (ii) is false.

Q6. (C or D). (i) is incorrect. A perfect square has any prime factor $2k$ time for some k . (ii) is incorrect. Since 0 has infinitely many positive factors. (Since even/odd property is not well-defined from $\aleph_0$, we consider C or D as both correct answers).

Q7. (B).

As the empty set is in D and, if there is a common element d of all three sets A, B, C then $\{d\}$ is also in D . Other subsets of A, B, C are not in D .

Q8. (C).

Note that B and the $C \cup D$ have the same number of elements and B has 2 elements more than C . Thus, D can have only 2 elements.

Q9. (A).

(B) is false since S_2 is not a subset of S_3 . (C) is false since S_1 does not equal to S_3 . (D) is false since S_3 is not a subset of S_4 . (E) is false since S_1 is not a subset of S_2 .

Q10. (B or D). $R = \{(true, true), (false, true), (false, false)\}$ Converse and inverse are equivalent.

Q12. (B). F has two elements $\{(0, 0), (1, 1)\}$, $\{(0, 0), (1, 1), (0, 1), (1, 0)\}$ and G has three elements $\{(0, 0), (1, 1)\}$, $\{(0, 0), (1, 1), (0, 1)\}$, $\{(0, 0), (1, 1), (1, 0)\}$.

Q14. (B). Reflexive relations will have all xRx elements.

Q15. (E). Statement II is false. Equivalence classes are members (not subsets).

Student Number: A

--	--	--	--	--	--	--	--

Part A: Multiple Choice Questions (Total: 30 marks)Please shade only ONE bubble for each question using **pencil**.

	(A)	(B)	(C)	(D)	(E)
1.	☐	☐	☐	☒	☐
2.	☐	☐	☐	☐	☒
3.	☐	☐	☐	☐	☒
4.	☐	☐	☒	☐	☐
5.	☐	☐	☒	☐	☐
6.	☐	☐	☒	☒	☐
7.	☐	☒	☐	☐	☐
8.	☐	☐	☒	☐	☐
9.	☒	☐	☐	☐	☐
10.	☐	☒	☐	☒	☐
11.	☐	☒	☐	☐	☐
12.	☐	☒	☐	☐	☐
13.	☐	☐	☐	☒	☐
14.	☐	☒	☐	☐	☐
15.	☐	☐	☐	☐	☒
	(A)	(B)	(C)	(D)	(E)

Part B (Total: 20 marks)

16. [6 marks]

Colour(s) of cat =

{grey}

Colour(s) of sheep =

{white}

Colour(s) of rabbit =

{brown, black}

Colour(s) of penguin =

{white, brown}

17. [14 marks]

(a) (i) $(R \circ S) \setminus (S \circ R) =$

 $\{(3, 6)\}$

[4]

(ii) $(R^{-1} \circ S^{-1}) \circ (S \circ R) =$

 $\{(6, 6), (9, 9)\}$

(b)

[2] (i) $A =$

 $\{2^n - 1 \mid n \in \mathbb{N}\}$ There are many possible answers.

(ii) $B =$

 $\{n \in \mathbb{N} \mid (3 \mid n) \wedge \text{odd}(n)\}$ There are many possible

(c)

[2]

(i) Is R_1 an equivalence relation?

Yes

If no, why?

(ii) Is R_2 an equivalence relation?

No

If no, why?

Not transitive.

(d)

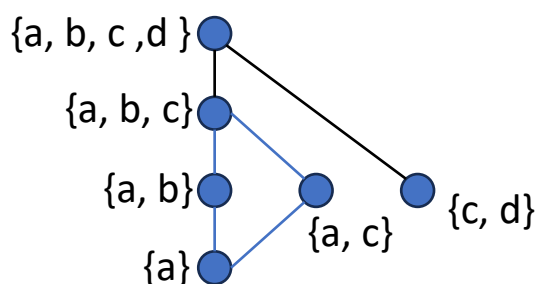
[2]

 $\mathbb{Z}^+ / \sim$ equals to $\{\{1\}\}$ or $\{\mathbb{Z}^+\}$ There are many possible answers.

(e)

[4]

(i) Hasse Diagram:



(ii) Is there a smallest element?

No

(iii) Are there any maximal elements?

Yes. $\{a, b, c, d\}$

=== END OF PAPER ===